FLEX+: moving the ceiling

An experiment branch. The ladder shape is fixed at six exits split at depth two, as the paper reports it; the question is whether the architectural ceiling above it can be raised without touching either.

1. What the ceiling is made of

The paper's ceiling is 38.33% counted. It is not a routing result and no better router can move it: it is what a frame costs when every tile already sits on the shallowest rung it is allowed. At a 0.2 dB budget the deployed system delivers 36.81% of that 38.33%, which is 96% of the ceiling, so the allocation is very nearly finished and the ceiling is the whole of what is left.

Decomposing that floor decides the experiment by arithmetic rather than by taste. A frame with every tile on the shallowest rung costs 0.6088 of a released decode, and one part of it is large enough to matter.

part of the floor	cost	share
shared trunk, 4 blocks, always, full frame	0.2981	49.0%
tiled trunk, 2 blocks	0.1491	24.5%
upsample, always, full frame	0.0816	13.4%
adapter at the shallowest exit	0.0464	7.6%
head, with its halo	0.0240	3.9%
seam repair	0.0095	1.6%

Table 1. Where the ceiling's floor goes. Frame-relative, halo on the head, from flexuf.cost on the paper's configuration.

Halving the shared trunk puts the ceiling at 54.0% and quartering it at 61.5%. Halving anything else leaves it below 47%: the upsample gone entirely would give 45.2%, the adapter 42.6%, the head 40.9%. There is one lever, and the target this branch was opened for -- 55% at a 0.2 dB budget -- is reachable only through it.

2. Why the exit ladder cannot reach it

The four blocks below the split run before the frame is tiled, on every pixel, whatever depth any tile chooses. Depth is one axis and it does not intersect them. Lowering the split depth would, and that is the move that is closed: it is also what creates the seam the paper spends a section on. So the ceiling needs a second axis, applied to the stem, that does not tile it.

2.1 What the literature offers

Width. Slimmable Networks (Yu et al., ICLR 2019) trains one network executable at several channel widths with shared weights and per-width normalisation, and reports each width close to a network trained at that width alone. Dynamic Slimmable Network (Li et al., CVPR 2021) makes the width input-dependent with a gate. SLEXNet (ACM TECS, 2024) combines slimmable widths with early exits, which is the pairing here. Cost falls with the square of the width.

Resolution. RANet (Yang et al., CVPR 2020) routes easy inputs through a low-resolution path and exits them early, keeping a high-resolution path for the rest. Half resolution is a quarter of the cost, the same scaling as half width.

Spatial sparsity. Dynamic Convolutions (Verelst and Tuytelaars, arXiv:1912.03203) learn a spatial mask and execute convolutions only where it is set; focused convolutions (arXiv:2310.07782) do the same for pretrained networks. Cost falls linearly with the active fraction, which is weaker, but it is the only one of the three that changes no tensor shape and therefore adds no tile boundary.

Width is the axis this branch follows, on instruction, and it is also the one with the strongest scaling. It is applied to the stem as a whole rather than per tile, so it creates no seam either.

3. The stem does not tolerate being cut

Before building anything, the cheapest question: can the stem simply be made smaller? Two training-free degradations, measured at the shallowest exit against the released decoder on twelve frames.

stem	q0	q32	q63
half	+4.27	+9.77	+13.94
drop 0.25	+0.50	+1.20	+2.21
drop 0.5	+1.14	+2.09	+3.29
drop 0.75	+1.96	+2.79	+4.07

Table 2. Extra dB from degrading the stem, untrained, above what the shallowest exit already costs. The budget this branch is aiming at is 0.2 dB in total.

Every entry is outside the budget, most of them by an order of magnitude. A stem at half resolution spends forty times the budget at the lowest rate and seventy at the highest. Dropping a quarter of the channels -- the mildest cut available -- costs two and a half times the whole allowance at the lowest rate and eleven times at the highest.

The deeper trunk repairs a little of the damage, and only a little: at a quarter of the channels dropped the deepest exit is 0.10 dB better off than the shallowest at the lowest rate and 0.15 dB at the highest, against a penalty of 0.50 and 2.21. It is not a mechanism to build on.

This is conclusive about cutting and says nothing about slimming. An untrained slice is exactly what the slimmable literature exists to improve on, so the result is a floor rather than an answer.

4. A narrow stem, trained to match the full one

The four stem blocks are replaced by a module of width w with a 1x1 projection either side, so everything downstream still sees the same channel count and needs no change. Its cost is w squared of the original plus the two projections. The residual is around the whole module and the output projection is zero-initialised, so at step zero it is exactly the decode that skips the stem -- a known starting point rather than noise.

It is trained to reproduce the full stem's output feature, with the encoder, the entropy model, the rest of the trunk, the exits and the head all frozen. The target is therefore a pure regression and any quality change is attributable to the stem alone.

objective	width	channels	ceiling	extra dB q0	q32	q63
decode	0.5	192	66.0%	+0.617	+1.571	+2.554
decode	0.707	272	63.4%	+0.452	+1.178	+1.992
feature	0.25	96	68.1%	-1.257	-2.373	-3.780
feature	0.35	136	67.4%	+0.572	+1.429	+2.379
feature	0.5	192	66.0%	-1.176	-2.260	-3.713
feature	0.707	272	63.4%	+0.405	+0.972	+1.665

Table 3. What each width costs and what it buys. Ceiling is hook-counted on the executed pass, the same counter the paper reports; extra dB is against the same decode with the full stem.

4.1 What the four widths say

Width buys very little. Across the whole sweep the compute in the stem rises eightfold, from a sixteenth of the original at w=0.25 to a half at w=0.707, and the distortion at the highest rate falls by about a third, from +2.57 to +1.67 dB. Every point is outside the 0.2 dB budget: the best of them, +0.41 dB at the lowest rate, is twice it, and the same width costs eight times it at the highest. The curve is not approaching the budget from above -- it is nearly flat in the direction that matters.

The ceiling hardly moves either, 68.1% down to 63.4%, because the stem is only half of the floor and the rest of it is unchanged whatever the stem costs. That is worth stating the other way round: the prize does not depend on getting the width small. Any stem replacement that met the budget would put the ceiling somewhere between 63 and 68%, well past the 55% this branch was opened for. The obstacle is fidelity, not cost.

Capacity is therefore not what binds. What binds is the objective. A 5% error in the feature comes out as 5.4 times the dB at the lowest rate and 9.6 at the highest, because the trunk below the split amplifies stem error rather than absorbing it -- and a mean squared error on the feature is indifferent to the direction of the error while the trunk is not. Matching the stem is the wrong thing to ask for; matching the decode is the thing that is measured.

Training does move the trade-off, and the size of the move is worth stating carefully. The trained half-width stem reaches about the distortion that untrained channel dropping reached while computing a quarter of the stem against three quarters of it -- but the dropping probe substitutes channels after running the whole stem, because it was written to ask what those channels are worth and not what skipping them saves. Its compute figure is therefore notional and the comparison is between one measured cost and one arithmetic one. The direction is not in doubt; the factor is.

4.2 The decode objective, and a comparison not to make

The first decode-objective run came out worse than the feature one at the same width, and the comparison is not usable. It trained for 12,000 steps at batch 4 against 20,000 at batch 8, and it also samples one exit per step out of four, so each exit saw roughly a sixth of the updates. The step count was reduced because a decode forward and backward is dearer than a feature one, which was a reasonable thing to want and an unreasonable way to get it: it traded away the only property that made the two runs comparable.

Matching the budget at 20,000 steps and batch 8 recovered most of the difference -- +2.51 to +1.99 dB at the highest rate -- and left the decode objective behind the feature one at all three rates, by ten to twenty per cent. The obvious suspect was the exit sampling: one exit of four per step gives each exit a quarter of the updates that the feature objective's single target gives all of them at once. Supervising every exit each step, at four times the cost, changed nothing -- +0.444, +1.207, +2.064 against +0.452, +1.178, +1.992 -- so that was not the reason.

The feature objective is simply the better one here, which is not what we expected of it: it optimises a proxy while the other optimises the reported quantity. The likely reason, and this is a hypothesis rather than a measurement, is dilution. The decode loss at an exit is the error that exit makes anyway plus the damage the narrow stem adds, and only the second part carries information about the stem; the feature target isolates the stem's contribution exactly. Optimising the measured quantity is not the better choice when the measured quantity is mostly something else.

None of this changes the answer to the question the branch asked. Across every width, objective and budget tried, the best point is +0.41 dB at the lowest rate and +1.67 at the highest, against a budget of 0.2. The width axis does not reach it on this decoder, and the remaining comparison decides only which objective is less far away.

4.3 The third axis, and what all three of them say

Spatial sparsity is the one axis left, and the cheapest to settle. Rank each position by how much the four stem blocks move it -- the norm of the stem's own residual -- and hand the least-moved fraction the stem's input instead of its output. The ranking is an oracle: it sees the output a learned mask would have to predict, so a learned mask cannot beat it and a failure here is conclusive.

active fraction	ceiling	+dB q0	q32	q63
1	39.1%	+0.000	+0.000	+0.000
0.75	46.6%	+0.491	+0.638	+0.960
0.5	54.0%	+1.047	+1.329	+2.011
0.25	61.5%	+1.605	+2.109	+3.250

Table 4. An oracle spatial mask on the stem. Quality measured; the ceiling is arithmetic, the active fraction times the stem's share of the floor. Nothing else in the floor moves.

Skipping the least-moved quarter of positions costs between +0.49 and +0.96 dB, two and a half to five times the budget. At half the positions -- the setting whose ceiling is the 54% this branch was aiming at -- it costs +1.05 to +2.01. The axis is closed, and closed harder than width, because no learned mask can do better than the ranking used here.

All three axes agree, and that is the result. Remove the stem's work by channel, by width or by position and the price is the same order: two to ten times the budget. The stem is not doing redundant work anywhere.

Set against the paper this is a sharp contrast worth stating. Content-adaptive *depth* works: the same decoder gives up 0.1 dB and returns 27.6% of its arithmetic, because some tiles genuinely need less of the trunk than others. Content-adaptive *stem* does not, on any of the three axes. The difference is what is being chosen. The exit ladder decides how much further to go from a shared representation; these probes remove the work that representation is made of. There are easy regions in a picture, and there are none in the computation that turns a latent into one.

4.4 The stem trained jointly with the ladder

Every width above was fitted to a frozen target: reproduce, with fewer channels, the representation a decoder trained without it produces. That is the hardest version of the task and the obvious objection to the negative result, so it was run the other way. The trunk and the adapters were unfrozen and trained together with the narrow stem for 20,000 steps, which lets the exits move to meet the stem rather than requiring the stem to reproduce something fitted without it.

width	training	ceiling	dB q0	q32	q63
0.5	frozen target	66.0%	0.585	1.330	2.162
0.5	trained jointly	66.0%	0.293	0.568	0.778
0.25	frozen target	68.1%	0.764	1.758	2.796
0.25	trained jointly	68.1%	0.305	0.575	0.779
1.0	the ladder as it ships	47.6%	0.108	0.171	0.225

Table 5. A narrow stem fitted to a frozen target against one trained with the ladder. Decibels below the released decoder at exit 2, over 12 sequences. The last row is the same exit with the stem the decoder ships, which is what the extra channels buy. The ceiling is unchanged by how the stem was trained: it is a property of the architecture.

Joint training helps, and by a lot: at width 0.5 the cost falls from 0.585 to 0.293 dB at q0 and from 2.162 to 0.778 at q63, so roughly half to a third of the damage was the frozen target rather than the missing channels. It is not enough. The cheapest point of the cheapest configuration is 0.293 dB, which is 1.5 times the 0.2 dB budget, and the highest rate costs 3.9 times it. Halving the stem again buys 2 points of ceiling and costs slightly more quality, which is the same flat trade the four widths showed.

The prediction recorded before this ran was that the frozen target was the obstacle. It was part of it, and the part it was is now measured; what remains is not. A stem with half the channels, trained with everything downstream free to accommodate it, still costs more than the budget the whole method is built around.

4.5 A router that can see a tile's neighbours

One more input, and it is not on the stem. Every group the paper's head reads is read at the tile: the bits pathway normalises the entropy coder's output by the frame mean, so a tile knows how it compares with the picture and not with what is next to it. Difficulty is spatially correlated -- that is why the exit maps have contiguous regions rather than salt and pepper -- so the neighbourhood mean is information the head does not have and could have for sixteen parameters. Two more channels on that pathway carry the rate box-filtered over a three-tile window and its log. They start at zero, the RNG stream is restored after the wider convolution is built, and the recipe is the one every variant of the paper's ablation runs: 3000 steps, lambda 1.3e-5, seed 0, on the pinned checkpoint. The only difference is what the head can see.

The run is not in results/ yet; this section prints its numbers when it is.

5. What this leaves

The branch set out to reach 55% at a 0.2 dB budget with the ladder shape left alone, and it does not get there. Every route to the only lever large enough -- the four always-on trunk blocks -- costs between two and ten times the budget, and one of the three routes was tested with an oracle, so it is not a matter of finding a better gate or a longer schedule.

One thing would still be worth running, and it is not a variation on what is here. The other -- training the narrow stem jointly with the ladder -- has now been run and is Section 4.4: it recovers half to two thirds of what the frozen target cost and still lands at 1.5 to 3.9 times the budget. Nothing here questioned the split depth, because the instruction was to leave it alone -- but the arithmetic in Section 1 says that lowering it is the one move left that could reach the target cleanly, at a seam cost this project has already measured (0.100/0.144/0.234 dB by split at 256 px, untrained). Between a known seam cost and three axes that each cost more, the seam is the cheaper problem.

The negative result is worth keeping either way. A decoder's shared stem is not slack: the ceiling in the paper is not an artefact of how the ladder was cut, and a reader who assumes there is easy work to remove below the split should be shown these numbers.